AI-R QUALITY: AI-BASED AIR QUALITY WARNING

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1. ENVIRONMENTAL PROBLEM

Air pollution is becoming a **serious public health challenge** across Vietnam, especially in **Ho Chi Minh City**. Although large amounts of environmental data are collected, they remain **difficult to access** and **interpret,** leaving both citizens and government agencies without clear, actionable insights for data-driven health decisions.

2. RESEARCH QUESTION

How can machine learning model using past **4 years data** to **classify and predict** the **trend of air quality** to **inform** people in Ho Chi Minh city and then **suggest** action to protect users' health?

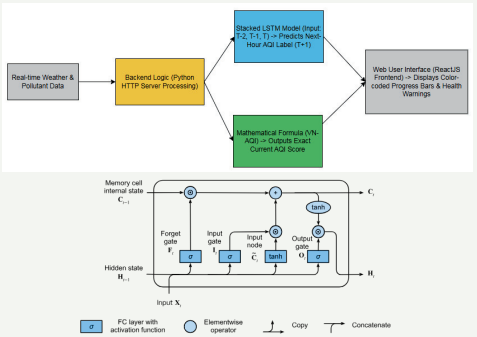
3. ENGINEERING GOALS

- Turn complex data into highly readable, standardized **5-tier severity index** using **AQI formula**
- Train AI to **forecast** upcoming air quality using LSTM model
- To give out **suitable recommended action** for users to protect their health

4. DATA & TOOLS

Data	35,064 samples from internet from 2022-2026
Parameters	CO ,NO ,O_3,SO_2,PM2.5 ,PM10 ,NH_3
Split	70% train/10% Validation/ 20% test
Tools	Python, TensorFlow, Keras, Google Colab

5. SYSTEM IDEA



AQI trend via Deep Learning

LSTM MODEL

6. AI MODEL & RISK SCORE

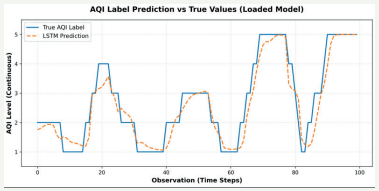
Designed to detect and learn temporal dependencies, trends, and seasonal patterns across the **35,064-sample** history.

Network Architecture: **Stacked Deep LSTM network** utilizing **3D Tensors** with the shape :

BatchSize=32 **TimeStep=3** **Features=16**

Optimization Layer: Utilizes **Dropout (0.2)** to **avoid overfitting**, **EarlyStopping** to **prevent overtraining**, and **ReduceLROnPlateau** for **automated**, highly **granular learning rate** adjustments during training.

7. SAMPLE RESULTS



REGRESSION METRICS

0.7962 R^2 Score (Accuracy)

0.29 MAE (Mean Absolute Error)

0.38 RMSE (Root Mean Squared Error)

8. PRODUCT DEMO



User interface that display **level of AQI** then based on the current **condition** and **forecasted AQI** to **give** out warning and recommended **action.**

9. ENVIRONMENTAL IMPACT

- **Supports early air pollution prediction** for proactive environmental management.
- **Improves public awareness** and contributes to healthier, more sustainable cities.

10. LIMITATIONS

- The current system **focuses** on air quality prediction **within the selected deployment region only**.
- **Extreme** environmental events may **affect prediction reliability.**
- The model can be further improved by **incorporating additional environmental and urban factors**

11. FUTURE WORK

- **Expand** the platform with **real-time environmental monitoring** and **multi-source data integration**.
- **Enhance prediction accuracy** using advanced AI models and deploy **GreenAir AI** as a scalable smart-city decision support system.